

# StaleyNET: Text-Conditioned Music Generation Trained from Scratch on Edge GPUs in Rust

Arpan Pathak

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**Abstract**

This paper describes StaleyNET, a small system that generates music from a short text description. We built it from first principles, mainly as a way of understanding what each part of such a system contributes. All components, including the GPU kernels, are written in Rust, and the models are trained from random initialisation on two NVIDIA Jetson devices with about 23 hours of Creative Commons music. Sound is represented by quantised log-mel spectrograms, and a decoder-only Transformer learns to predict these tokens from a caption and, optionally, from lyrics. Because every stage is our own, we could examine the stages one at a time. These measurements suggest that the quality of the generated audio is limited mainly by the tokenizer rather than by the language model: a model trained on a single excerpt reproduces it exactly, while the tokenizer on its own loses much of the spectral detail. Guided by this observation, we revised the tokenizer and roughly halved its reconstruction error. We also describe a simple paired test for whether a model makes use of the lyrics it is given; our first lyrics-conditioned models do not yet pass it. Each algorithm is presented with a small worked example, in the hope that the paper is useful to readers who would like to build a similar system.

## 1 Introduction

Generating music from a sentence of text has made remarkable progress in recent years [1, 3]. Most current systems combine several large components that are trained separately: a neural codec that turns audio into tokens, a text encoder that turns the prompt into vectors, and a Transformer that relates the two. This division of labour works very well, although it can make it difficult to tell which component is responsible for a particular quality, or a particular flaw, in what one hears.

In this work we approached the problem from the other direction and built a complete, if modest, system ourselves (Figure 1). StaleyNET computes its spectrograms, learns its tokenizer, trains its Transformer and reconstructs its waveforms without any pretrained component. It is written in Rust, and its GPU kernels are compiled from Rust by cuda-oxide [19]. We trained it

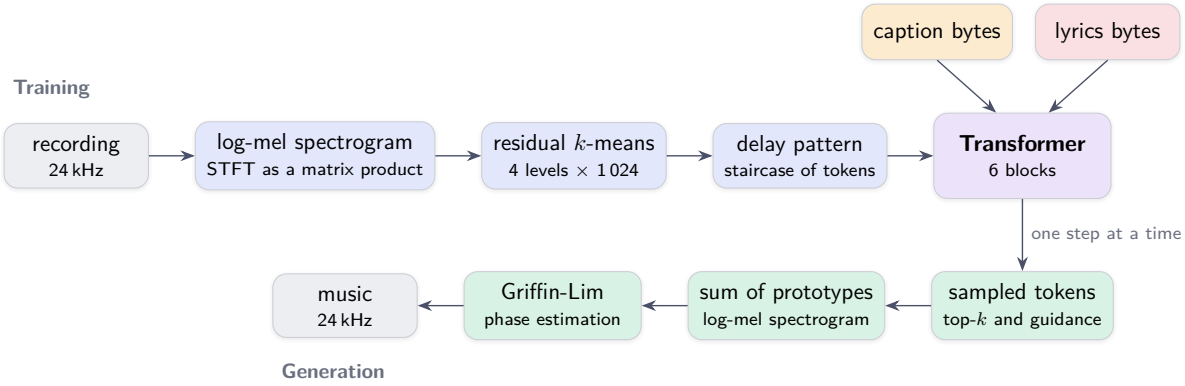



Figure 1: An overview of StaleyNET. During training (top row), a recording is converted into a staircase of discrete tokens, which the Transformer learns to predict from the caption and the lyrics. During generation (bottom row), the Transformer produces tokens one step at a time, and the tokens are turned back into sound.

on an NVIDIA Jetson Orin NX with 16 GB of memory and on an NVIDIA Jetson AGX Thor with 128 GB, two embedded computers designed for robotics. Owning every stage gave us the opportunity to study each one in isolation, and much of the paper is about what that study revealed.

Our first models learned text-conditioned structure well above chance, yet the voices they produced sounded blurred. Two experiments helped us locate the cause. When we passed held-out recordings through the tokenizer and the vocoder alone, without any model, most of the spectral detail was already lost. When we trained the model on a single excerpt, it reproduced that excerpt token for token. Together these results suggested that the tokenizer was the main limitation, and a revised tokenizer that quantises groups of frequency bands separately reduced the reconstruction error by about half.

The contributions of this paper are as follows.

- A complete text-to-music pipeline written in Rust, including its GPU kernels, and trained from scratch on embedded GPUs.
- A description of each algorithm with a worked example and, where an algorithm is easy to get wrong, an argument for its correctness.
- A way of adding a lyrics input to a trained model without discarding what it has learned, together with a paired test of whether the input is used.
- A stage-by-stage analysis that separates the errors of the tokenizer, the vocoder and the language model, and a tokenizer study guided by it.

We write algorithms in the style of Cormen et al. [4], as in our earlier tutorial on tiled matrix multiplication [20]. Each procedure first states its inputs and its output in words, with their units; its lines are numbered, and the text refers to them. The range $[a, b)$ denotes the integers $a, \ldots, b-1$, and indices start at 0. Every symbol is defined where it first appears, and Table 1 collects the ones used in more than one section. A few letters are reused for a different purpose inside a single subsection, for example $K$ for the keys of attention; the text says so where this happens.

Two units recur throughout. Frequencies and sample rates are given in hertz (Hz): a tone of 440 Hz vibrates 440 times per second, and a sample rate of 24 kHz means that the waveform is recorded as 24 000 numbers per second. Losses are given in *nats*, the unit of information obtained with the natural logarithm: a prediction that gives the correct answer probability $q$

costs $-\ln q$ nats, so a certain prediction costs 0 and a uniform guess among 1 024 codes costs $\ln 1024 \approx 6.93$.

Table 1: Symbols used in more than one section.

| symbol | meaning | value |
|---|---|---|
| $f_s$ | sample rate of the analysed audio (samples per second) | 24 000 Hz |
| $N$ | samples in one spectrogram frame | 1 024 (42.7 ms) |
| $H$ | hop: samples between the starts of two frames | 256 (10.7 ms) |
| $S$ | log-mel spectrogram, one row per frame, one column per mel band | $\cdot \times 128$ |
| step | the frames described by one set of codes | 1 or 2 frames |
| $C$ | codebooks (codes) per step | 4 or 16 |
| $K$ | prototypes per codebook | 1 024 |
| $G$ | groups of mel bands quantised separately | 1 or 4 |
| $d$ | width of the vectors inside the model | 256 or 512 |
| $L$ | bytes reserved for lyrics | 256 |
| $\theta$ | all trainable parameters of the model | 7.1 M or 42.9 M |

## 2 Related work

AudioLM [2] showed that audio can be generated by a language model over discrete tokens. MusicLM [1] conditioned this approach on text, and MusicGen [3] simplified it to a single Transformer over the codebooks of a neural codec, interleaved by a delay pattern. We follow MusicGen in using the delay pattern, top-$k$ sampling [9] and classifier-free guidance [12], and we replace its pretrained components with $k$-means quantisation of log-mel frames and with raw text bytes, as studied for text by ByT5 [31].

Jukebox [7] generated singing from lyrics and learned the correspondence between words and sound from 1.2 million songs. Singing synthesis systems such as DiffSinger [16] instead work from a musical score with phoneme durations. Our lyrics experiments rely on the line-level timings of the JamendoLyrics benchmark [27, 8]. Neural codecs [32, 6, 15] and neural vocoders [14, 24] achieve far higher fidelity than $k$-means tokens and the Griffin-Lim algorithm [10, 21], at a training cost beyond our setting; Section 9.1 measures what our simpler choices cost.

## 3 From sound to spectrogram and back

A recording is stored as a *waveform*: a list of *samples*, each the displacement of the loudspeaker at one instant, scaled to lie between $-1$ and 1. The *sample rate* is the number of samples per second, measured in hertz, and a rate of 24 kHz therefore gives 24 000 numbers per second. Any sound can be described as a sum of vibrations, each with a *frequency* (how many times per second it repeats, in Hz), a *magnitude* (how strong it is) and a *phase* (where in its cycle it is at a given instant). The ear is sensitive to frequency and magnitude but much less to phase. We therefore describe sound by its log-mel spectrogram: the energy in 128 frequency bands spaced according to human pitch perception, measured $f_s/H = 24\,000/256 = 93.75$ times per second and compressed logarithmically.

### 3.1 Resampling

The training recordings arrive at several sample rates, most often 44.1 kHz (the rate of a CD), and we convert them all to 24 kHz. A waveform sampled at a rate $f$ can only represent

frequencies below $f/2$, the *Nyquist frequency*, because at least two samples per cycle are needed to tell a vibration apart from silence. Lowering the rate to 24 kHz therefore requires first removing all content above 12 kHz; otherwise that content folds back into the audible range as false tones, an effect known as *aliasing*.

**Definition 3.1** (Bandlimited resampling)**.** Let $x_m$ be sample $m$ of a recording made at $f_{\text{in}}$ samples per second, and let $f_{\text{out}}$ be the desired rate, also in samples per second. Output sample $n$ describes the instant $n/f_{\text{out}}$ seconds, which lies

$$t_n = n\,\frac{f_{\text{in}}}{f_{\text{out}}}$$

input samples after the start; $t_n$ is usually not a whole number. The resampled signal is

$$y_n \;=\; \sum_{m} x_m\, h(t_n - m)\,,$$

where the *filter* $h$ gives the weight of an input sample as a function of its distance $t_n - m$ from $t_n$, measured in input samples. Each output sample is therefore a weighted sum of the input samples near $t_n$.

The ideal weights come from the function $\operatorname{sinc}(u) = \sin(\pi u)/(\pi u)$, which interpolates a bandlimited signal exactly but extends infinitely far in both directions [26]. We use the windowed sinc filter

$$h(t) = c\,\operatorname{sinc}(ct)\,w(ct/Z),$$

whose three parameters are as follows. The *cutoff* $c$, a number between 0 and 1, is the highest frequency kept, as a fraction of the input Nyquist frequency; when the rate is lowered, $c$ is slightly below $f_{\text{out}}/f_{\text{in}}$. $Z = 24$ is the number of zero crossings of the sinc kept on each side, which sets how many input samples contribute to each output sample. $w$ is a Kaiser window, a bell-shaped taper that falls smoothly to zero at $\pm 1$ so that cutting the sinc short does not cause ripples; its shape parameter $\beta = 8.6$ trades a sharper cutoff against stronger suppression of the removed frequencies. Evaluating $h$ anew for every output sample would be slow, but fortunately the fractional part of $t_n$ takes only a few values.

**A worked example.** For a conversion from $f_{\text{in}} = 44\,100$ Hz to $f_{\text{out}} = 24\,000$ Hz, the greatest common divisor of the two rates is 300, so $t_n = n \cdot 44\,100/24\,000 = 147n/80$ input samples. For $n = 1$ we obtain $t_1 = 1 + 67/80$: the second output sample lies $67/80$ of the way from input sample 1 to input sample 2. Every $t_n$ has a fractional part of the form $p/80$ with $p \in [0, 80)$, so exactly 80 sets of filter weights are needed. These sets are called the *phases* of a polyphase filter, and they are computed once.

**RESAMPLE**$(x, f_{\text{in}}, f_{\text{out}})$

*Input:* the samples $x_0, x_1, \ldots$ of a recording (numbers between $-1$ and 1), the rate $f_{\text{in}}$ at which they were recorded, and the desired rate $f_{\text{out}}$; both rates are whole numbers of samples per second (Hz).
*Output:* the samples $y_0, y_1, \ldots$ of the same sound at $f_{\text{out}}$ samples per second; one second of input gives $f_{\text{out}}$ output samples.

```
1   g = gcd(f_in, f_out)
2   L = f_out / g   // number of phases
3   M = f_in / g   // input step per output sample, in L-ths
4   c = 0.97 · min(1, f_out / f_in)   // cutoff
5   R = ⌈Z / c⌉   // filter half-width
6   for p ∈ [0, L) {
7       for j ∈ [−R, R] {
8           h_p[j] = c sinc(c (j − p/L)) w(c (j − p/L) / Z)
9       }
10  }
11  for each output index n {
12      m = ⌊nM / L⌋   // whole part of t_n
13      p = nM mod L   // fraction of t_n
14      y_n = Σ_{j∈[−R,R]} h_p[j] x_{m+j}
15  }
```

Lines 1–5 determine the geometry of the filter. $g$ is the greatest common divisor of the two rates, $L$ is the number of distinct fractional positions (phases), and $M/L = f_{\text{in}}/f_{\text{out}}$ is the distance between consecutive output samples, measured in input samples. $R$ is the number of input samples used on each side of $t_n$; it grows as $c$ shrinks, because a lower cutoff needs a wider filter. Lines 6–10 compute the $L$ phases once, each a list of $2R + 1$ weights $h_p[j]$, and line 14 is the single dot product needed for each output sample. The factor 0.97 in line 4 places the cutoff at 97 % of the new Nyquist frequency, so that the filter has a narrow band in which to fall from full strength to zero before aliasing could begin. A test confirms that the filter does its job: when 48 kHz audio is converted to 24 kHz, a 15 kHz tone, which the new rate cannot represent, is attenuated by more than 80 dB, a factor of $10^8$ in power.

### 3.2 The spectrogram as a matrix product

The signal is divided into overlapping *frames* of $N = 1024$ samples (42.7 ms at 24 kHz), a new frame starting every $H = 256$ samples (10.7 ms). Each frame is multiplied by a Hann window, $w[n] = \sin^2(\pi n/N)$, which fades the frame in and out so that its abrupt edges do not appear as spurious frequencies. The strength of each frequency in each frame is usually computed with the fast Fourier transform (FFT). We computed it instead with a matrix product, so that the spectrogram runs on the same GPU kernel as the Transformer and benefits from the same tests (Section 8).

**Definition 3.2** (Log-mel spectrogram)**.** Let $F$ be the matrix whose row $f$ holds the $N$ samples of frame $f$, and let $W$ be the $N \times (N/2+1)$ matrix with entries $W_{nk} = w[n]\, e^{-2\pi i k n/N}$, where $n \in [0, N)$ indexes the samples of a frame, $k \in [0, N/2]$ indexes the frequencies, $w$ is the Hann window and $i = \sqrt{-1}$. The log-mel spectrogram is

$$S \;=\; \ln \max\bigl(|FW|^2 B,\; 10^{-5}\bigr),$$

where the absolute value and the square are taken entry by entry, and $B$ is the $(N/2+1) \times 128$ matrix whose columns are 128 triangular mel filters [25]. $S$ has one row per frame and one column per mel band, and its entries are natural logarithms of energy.

Column $k$ of $W$ holds a windowed cosine (real part) and sine (imaginary part) at frequency $k f_\mathrm{s}/N = k \cdot 23.4\,\mathrm{Hz}$, so entry $(f, k)$ of $FW$ measures how much of that frequency frame $f$ contains, and its square $|FW|^2$ is the energy at that frequency. The mel filters then pool these 513 frequencies into 128 bands. The *mel* scale follows pitch perception: it is linear below 1 kHz and logarithmic above, so the centres of neighbouring bands are about 26 Hz apart at low frequencies and about 330 Hz apart near 12 kHz, much as the ear distinguishes low notes more finely than high ones. The logarithm reflects the roughly logarithmic perception of loudness, and the floor $10^{-5}$ keeps silence finite. The real and imaginary parts of $W$ are two real matrices, and $B$ adds a third product. This formulation needs about $N/\log_2 N \approx 100$ times the arithmetic of an FFT, because a matrix product computes each frequency separately, whereas the FFT shares work between frequencies. Even so, on the Orin NX it analyses 60 s of audio in 47 ms, against 73 ms for a multithreaded CPU FFT (Table 9), and it agrees with the FFT to within $10^{-2}$ log units.

### 3.3 Estimating the phase

A generated spectrogram tells us how strong each frequency is, but not its phase, and a waveform cannot be built without one. Not every pattern of magnitudes and phases belongs to a real waveform: neighbouring frames overlap by three quarters, so they must agree about the samples they share. The Griffin-Lim algorithm [10] estimates a phase by alternating between two requirements: the magnitudes should match the target, and the spectrogram should be that of an actual waveform (it should be *consistent*).

**GRIFFIN-LIM**( $|S|, T$ )

*Input:* the target magnitudes $|S|$, a matrix of non-negative numbers with one row per frame and one column for each of the $N/2 + 1 = 513$ frequencies, and the number of rounds $T$.
*Output:* a waveform $x$ at $f_s$ samples per second whose spectrogram approximates $|S|$.

1 $X = |S| \cdot e^{i\phi}$ with random phases $\phi$
2 $x = \mathrm{ISTFT}(X)$
3 $P_{\mathrm{prev}} = \mathrm{STFT}(x)$
4 **for** $t \in [0, T)$ {
5 $P = \mathrm{STFT}(x)$ *// nearest consistent spectrogram*
6 $A = P + \alpha\,(P - P_{\mathrm{prev}})$ *// momentum*
7 $X = |S| \cdot A/|A|$ *// keep the phase, restore the magnitude*
8 $P_{\mathrm{prev}} = P$
9 $x = \mathrm{ISTFT}(X)$
10 }
11 **return** $x$

Here STFT (short-time Fourier transform) turns a waveform into its complex spectrogram, $FW$ in the notation above, and ISTFT turns a complex spectrogram back into a waveform by overlapping and adding its frames. Each entry of a complex spectrogram is a magnitude times $e^{i\phi}$, where $\phi$ is the phase in radians. In line 1, $X$ combines the target magnitudes with random phases. Line 5 replaces the current estimate by the spectrogram $P$ of a real waveform, and line 7 keeps the phase of each entry of $A$ while restoring its target magnitude. Line 6 is the acceleration proposed by Perraudin et al. [21], which extrapolates along the last change with momentum $\alpha = 0.99$; in the first round it has no effect, because $P = P_{\mathrm{prev}}$. We use $T = 64$ rounds. Because $S$ holds logarithms of mel energies, the target magnitudes are obtained by undoing each step of the definition: $e^S$ gives the band energies, a regularised least-squares inverse of $B$ spreads them back over the 513 frequencies, and a square root turns energy into magnitude.

# 4 From spectrogram to tokens

A language model chooses each next symbol from a finite vocabulary, whereas a spectrogram consists of real numbers. The tokenizer bridges this gap by replacing each short stretch of spectrogram with a few integers, its *codes*. Each band of the spectrogram is first standardised: its mean over the training set is subtracted and the result is divided by its standard deviation, so that quiet and loud bands carry similar weight and every value is a plain number without unit. A *step* is then formed from consecutive frames; in our first tokenizer a step joins 2 frames, giving a vector of $D = 2 \cdot 128 = 256$ numbers that covers $2H/f_s = 21.3\,\mathrm{ms}$ of sound.

## 4.1 Residual quantisation

One can think of residual quantisation as describing a colour first by its family, then by its shade within the family, and then by finer and finer corrections. Each level names the closest of $K$ prototypes to whatever the previous levels have left unexplained. A *prototype* is a typical step, a vector of $D$ numbers, and a *codebook* is a list of $K$ prototypes; the code chosen at a level is the position of the chosen prototype in its list, an integer in $[0, K)$.

**Definition 4.1** (Residual vector quantisation)**.** Let $E_0, \dots, E_{C-1}$ be $C$ codebooks (one per level $\ell$), each a list of $K$ prototype vectors of length $D$, and write $E_\ell[k]$ for prototype $k$ of codebook $\ell$. A step $x \in \mathbb{R}^D$ is encoded as the codes $c_0, \dots, c_{C-1}$ defined by

$$r_0 = x, \qquad c_\ell = \operatorname*{arg\,min}_{k \in [0,K)} \|r_\ell - E_\ell[k]\|^2,$$
$$r_{\ell+1} = r_\ell - E_\ell[c_\ell],$$

and it is decoded as $\hat{x} = \sum_\ell E_\ell[c_\ell]$. The vector $r_\ell$ is the *residual*: the part of $x$ that levels $0, \dots, \ell - 1$ have not yet described.

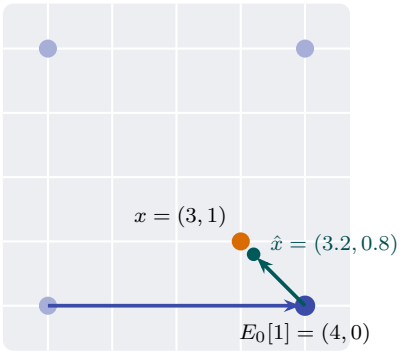



Figure 2: Residual quantisation of $x = (3, 1)$ in two levels. Level 0 (blue) chooses the prototype $(4, 0)$, and level 1 (green) adds the correction $(-0.8, 0.8)$.

**A worked example.** In two dimensions, consider the codebooks

$$E_0 = \{(0,0),\ (4,0),\ (0,4),\ (4,4)\},$$
$$E_1 = \{(0.8,0.8),\ (-0.8,0.8),\ (-0.8,-0.8),\ (0.8,-0.8)\},$$

and the point $x = (3, 1)$ (Figure 2). The squared distances from $x$ to the prototypes of $E_0$ are $10, 2, 18, 10$, so $c_0 = 1$ and $r_1 = (-1, 1)$. The closest entry of $E_1$ to $(-1, 1)$ is $(-0.8, 0.8)$, so $c_1 = 1$ and $r_2 = (-0.2, 0.2)$. The error falls from $\sqrt{2} \approx 1.41$ after one level to $\sqrt{0.08} \approx 0.28$ after two.

## 4.2 Training the codebooks

**TRAIN-CODEBOOKS**$(X, C, K, I)$

*Input:* a matrix $X$ with one row per training step and $D$ standardised values per row, the number of levels $C$, the number of prototypes per level $K$, and the number of refinement rounds $I$.
*Output:* the codebooks $E_0, \dots, E_{C-1}$, each a $K \times D$ matrix whose rows are prototypes.

```
1   R = X // what remains to be described
2   for ℓ ∈ [0, C) {
3       E_ℓ = K rows of R, chosen at random
4       repeat I times {
5           a_i = NEAREST(R_i, E_ℓ) for each row i
6           E_ℓ[k] = mean{R_i : a_i = k} for each k
7           if no row has a_i = k
8               E_ℓ[k] = a random row of R
9       }
10      R_i = R_i − E_ℓ[NEAREST(R_i, E_ℓ)] for each row i
11  }
```

**NEAREST**$(r, E)$ returns the index of the prototype in $E$ closest to $r$ in Euclidean distance. $R$ holds the residuals of all training steps, one row per step, and $a_i \in [0, K)$ records which prototype row $i$ is currently assigned to. Lines 5–6 are the two moves of Lloyd's algorithm [17]: each row is assigned to its nearest prototype, and each prototype then moves to the mean of its rows. Lines 7–8 restart a prototype to which no row was assigned, so that none of the $K$ entries goes unused, and line 10 passes the residuals on to the next level. We use $I = 20$ rounds on 200 000 steps drawn from the training set.

## 4.3 The nearest prototype as a matrix product

Finding the nearest prototype for every step is the most expensive part of both training and encoding. The following identity turns it into a matrix product, which the GPU computes efficiently.

**Lemma 4.1.** *For a vector $x$ and prototypes $e_0, \dots, e_{K-1}$, all of length $D$,* $\arg\min_k \|x - e_k\|^2 = \arg\min_k \left(\|e_k\|^2 - 2\, x \cdot e_k\right)$.

*Proof.* Expanding gives $\|x - e_k\|^2 = \|x\|^2 - 2\, x \cdot e_k + \|e_k\|^2$, and the first term does not depend on $k$. □

In the example above, the squared norms $\|e_k\|^2$ are $0, 16, 16, 32$ and the dot products $x \cdot e_k$ are $0, 12, 4, 16$, so the scores are $0, -8, 8, 0$, and the smallest again selects $(4, 0)$. On the GPU the dot products of 8 192 steps with 1 024 prototypes form a single matrix product, after which a small kernel finds the smallest score in each row.

## 4.4 Quantising groups of bands

The analysis of Section 9.1 led us to a refinement. Instead of quantising a whole step at once, we divide its 128 bands into $G$ equal groups of neighbouring bands, from the lowest frequencies

to the highest, and give each group its own residual codebooks. With one frame per step and $G = 4$, each group is a vector of $128/4 = 32$ numbers, which $K = 1\,024$ prototypes can describe more finely than a whole step of 128. In our experiments the grouping helped somewhat less than we expected, and the additional codes helped more (Appendix C). The codes of a step are ordered level by level: first the coarse codes of all $G$ groups, then their first corrections, and so on, so that the coarse-to-fine order relied on by the delay pattern (Section 5) is preserved. With $G = 4$ groups and 4 levels each, a step is described by 16 codes. Table 10 in the appendix compares this and other configurations.

# 5 The model

The model (Figure 3) is a decoder-only Transformer [28, 23]. It reads text and audio as a single sequence and predicts, at each position, what comes next, much as a text model predicts the next word.

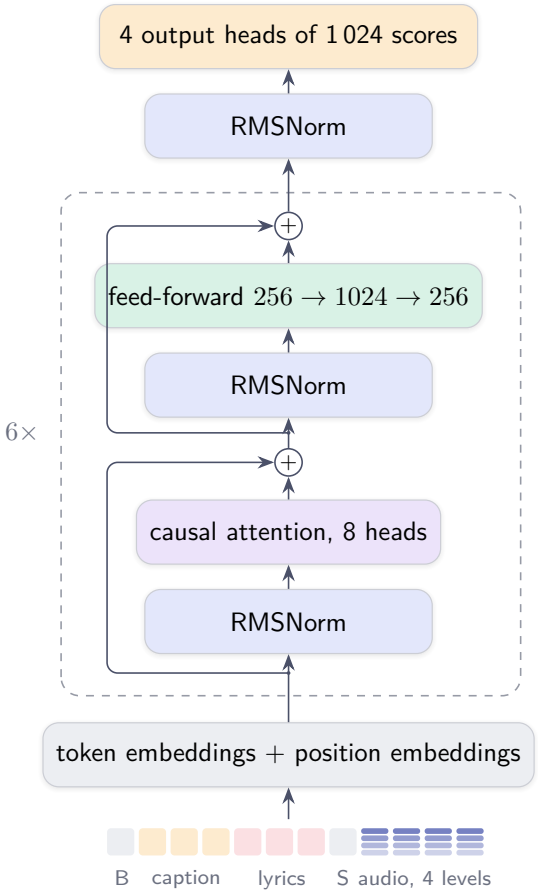



Figure 3: The model. Text bytes and audio tokens share one sequence, and each audio position carries all the codes of one step. A stack of pre-normalised blocks, each adding its result to its input, ends in one output head per code.

## 5.1 One sequence for text and audio

The model reads a sequence of *positions*, and at each position it receives one or more symbols from a fixed vocabulary. The vocabulary contains the 256 possible byte values of UTF-8 text [31], three markers (`BEGIN`, `SEPARATOR` and `PADDING`) and $C \cdot K$ audio symbols, one for each

prototype of each codebook. The caption is given as raw bytes, so no separate text encoder is needed. The text occupies a fixed 128 positions, namely `BEGIN`, up to 126 caption bytes (padded when shorter) and `SEPARATOR`, so that the audio always begins at the same position. An audio position carries all $C$ codes of one step. Every symbol and every position has an *embedding*, a learned vector of $d$ numbers, and the input vector of an audio position is the sum of the embeddings of its $C$ codes and of the position.

## 5.2 The delay pattern

At each position the model predicts all $C$ codes of a step. A finer code, however, corrects a coarser one and is easier to predict once the coarser one is known.

**Definition 5.1** (Delay pattern [3])**.** Write $a_u^\ell$ for the code of codebook $\ell$ at step $u$. Position $t$ of the delayed sequence holds, in slot $\ell \in [0, C)$, the code $a_{t-\ell}^\ell$, so that codebook $\ell$ runs $\ell$ positions behind codebook 0, and an empty marker when $t - \ell$ is not a step of the piece. A piece of $F$ steps becomes $F + C - 1$ positions.

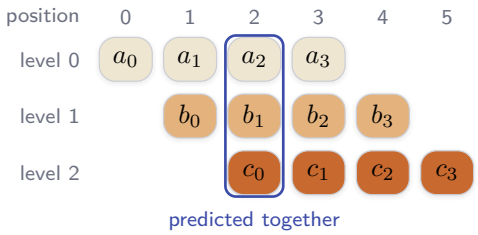



Figure 4: The delay pattern for three codebooks and four steps. The letters $a$, $b$ and $c$ stand for codebooks 0, 1 and 2, and the subscript is the step. At position 2 the model predicts $a_2$, $b_1$ and $c_0$, and the coarser codes that they refine are already part of its input.

In Figure 4, position 2 predicts $a_2$, $b_1$ and $c_0$ together. By then $a_1$, which $b_1$ refines, and $a_0$ and $b_0$, which $c_0$ refines, are already known. The price is $C - 1$ additional positions, whereas giving each codebook its own positions would multiply the length of the sequence by $C$.

## 5.3 The network and its objective

In each block, attention lets every position gather information from itself and from earlier positions, and a small feed-forward network then transforms each position on its own. Both parts normalise their input [33, 30] and add their result to it:

$$h = x + \text{Attention}(\text{Norm}(x)),$$
$$y = h + \text{FeedForward}(\text{Norm}(h)),$$

where $x$ is the sequence entering the block, one vector of $d$ numbers per position, and $y$ the sequence leaving it, of the same size. Norm rescales each vector to unit root-mean-square size and multiplies it by learned weights, and attention is split into several *heads*, each of which attends with its own $d/\text{heads}$ numbers. The feed-forward network expands each vector to $4d$ numbers, applies the GELU activation [11], and projects it back to $d$. Our first model has 6 blocks of width $d = 256$ with 8 attention heads (7.1 million parameters); the larger model of Section 9.3 has 8 blocks of width $d = 512$ (42.9 million parameters).

At position $p$ the model outputs, for each codebook $\ell$, a score for each of the $K$ possible codes, and a softmax turns the scores into a probability distribution over the code $y_{p,\ell}$ that

appears at position $p+1$. The loss is the cross-entropy, averaged first over the set $V_\ell$ of positions that have a target in codebook $\ell$ and then over the codebooks:

$$\mathcal{L} = \frac{1}{C} \sum_{\ell=0}^{C-1} \frac{1}{|V_\ell|} \sum_{p \in V_\ell} -\ln P_\theta(y_{p,\ell} \mid z_0, \ldots, z_p),$$

where $z_p$ is the input at position $p$, $\theta$ denotes the parameters, and $P_\theta(y \mid \cdot)$ is the probability that the model assigns to the correct code $y$. The loss is measured in nats per code. A model that assigns equal probability to all 1 024 codes has a loss of $\ln 1024 \approx 6.93$, and lower values mean that the model has learned something about the music.

### 5.4 Gradients by hand

As far as we know, no automatic differentiation is available for Rust on the GPU, so each layer implements its own gradient. Attention is the least obvious case. For one head, let the rows of $Q$, $K$ and $V$ be the queries, keys and values of the positions: each position *asks* with its query, the similarity of a query and a key decides how much attention a position pays to another, and the values are what is passed along. In this subsection $K$ and $V$ denote keys and values rather than codebook sizes, and $d$ denotes the width of one head. With $T$ positions, $Q$, $K$ and $V$ are $T \times d$ matrices, and

$$S = QK^\top/\sqrt{d} + M, \qquad P = \operatorname{softmax}(S), \qquad O = PV,$$

where $S$, $P$ and the mask $M$ are $T \times T$, $O$ is $T \times d$, $M$ is $-\infty$ above the diagonal so that no position attends to a later one, and the softmax turns each row of scores into weights that are positive and sum to 1.

**Proposition 5.1.** *Let $\partial O$ be the gradient of the loss with respect to $O$, and similarly for the other matrices, and let $\delta_{jk}$ be 1 when $j = k$ and 0 otherwise. Then $\partial V = P^\top \partial O$, $\partial P = \partial O\, V^\top$, $\partial S_{ik} = P_{ik}(\partial P_{ik} - \rho_i)$ with $\rho_i = \sum_j \partial P_{ij} P_{ij}$, $\partial Q = \partial S\, K/\sqrt{d}$ and $\partial K = \partial S^\top Q/\sqrt{d}$.*

*Proof.* The first two identities follow from $O = PV$. Within a row, $\partial P_{ij}/\partial S_{ik} = P_{ij}(\delta_{jk} - P_{ik})$, so $\partial S_{ik} = \sum_j \partial P_{ij} P_{ij}(\delta_{jk} - P_{ik}) = P_{ik}(\partial P_{ik} - \rho_i)$. The last two follow from $S = QK^\top/\sqrt{d}$. Where $M = -\infty$ we have $P_{ik} = 0$ and therefore $\partial S_{ik} = 0$. □

Apart from $\partial S$, every gradient is a matrix product and runs on the same kernel as the forward pass. Because hand-written gradients are easy to get wrong, a test perturbs three random entries of every parameter tensor by $\pm 10^{-2}$ and checks that the resulting change of the loss agrees with the computed gradient to within 5 %.

## 6 Lyrics

A caption describes how a piece should sound, but not which words should be sung. To make the words controllable, the model needs a second text input, and it needs training examples in which it can hear which words are sung where.

### 6.1 A position map that preserves a trained model

The lyrics occupy $L = 256$ byte positions between the caption and the separator. Each position $p$ takes its position embedding from row $\pi(p)$ of a table of learned vectors, and the choice of

the map $\pi$ determines whether a model trained without lyrics can later be extended (Figure 5). Simply inserting the lyrics would shift every later position, so the audio would read embeddings that the trained model had learned for other places.

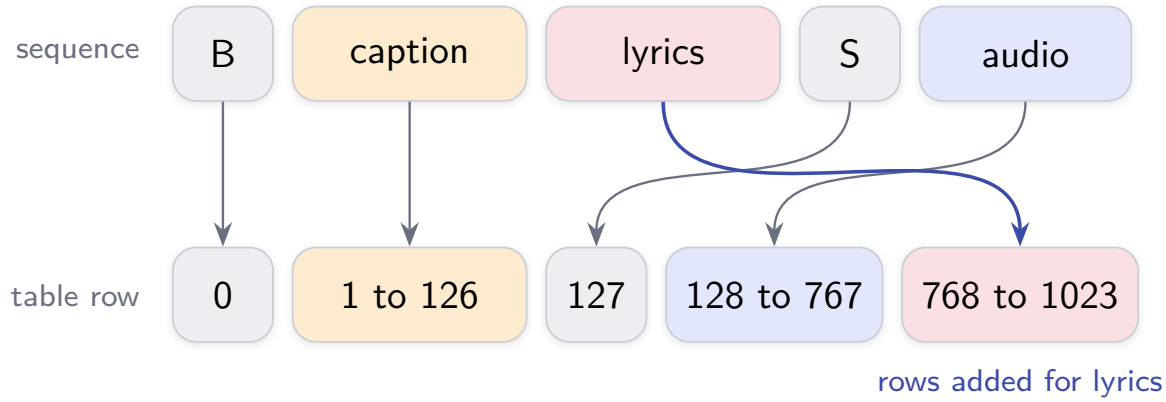



Figure 5: The position map $\pi$. Positions outside the lyrics keep the rows that they used before the lyrics were added, and the lyrics use new rows appended to the table.

**Definition 6.1** (Position map)**.** Suppose that position 0 holds `BEGIN`, the caption occupies positions $[1, 1{+}P)$, where $P$ is the number of caption bytes, the lyrics occupy $[1{+}P, 1{+}P{+}L)$, and the position table of the model without lyrics has $T$ rows, one for each position it can read. Then

$$\pi(p) = \begin{cases} p & \text{if } p < 1 + P, \\ T + p - 1 - P & \text{if } 1 + P \le p < 1 + P + L, \\ p - L & \text{otherwise.} \end{cases}$$

**Lemma 6.1.** *Every position outside the lyrics uses the row that the same part of the sequence used without lyrics, and the lyrics use only the new rows* $[T, T + L)$.

*Proof.* Caption positions are unchanged. The separator and the audio are shifted right by $L$ positions, and $\pi$ subtracts $L$. The lyrics are mapped onto $T + [0, L)$. □

For example, with $P = 2$, $L = 3$ and $T = 16$, positions 0 to 8 use the rows

$$0,\ 1,\ 2,\ 16,\ 17,\ 18,\ 3,\ 4,\ 5.$$

A trained model can therefore be extended by appending $L$ new rows to its table. All learned parameters keep their meaning, and training simply continues.

### 6.2 Pairing audio with the words sung in it

A training example covers a window of a song, and it should carry the lines sung in that window. JamendoLyrics [8] gives the start and end time of every line of 78 songs in English, French, German and Spanish, and a window, the stretch of a song shown to the model in one training example, receives each line whose time span overlaps it. Takes recorded with our studio come with their text but without timings, so we assume that their lines are spread evenly over the take; this is a fair approximation for short takes of continuous singing. Because the lyrics corpus is small (4.7 hours, against 23 hours of captioned music), each dataset receives a sampling weight: a weight of $w$ makes each of its songs $w$ times as likely to be drawn.

### 6.3 Testing whether the lyrics are used

A model can lower its loss on the lyrics corpus simply by learning how those particular songs sound, without reading their words. We therefore test the use of the lyrics directly, by giving the model the same audio with the correct words, with no words, and with words from another song.

**LYRICS-TEST**$(\theta, \mathcal{D}, B)$

*Input:* a trained model with parameters $\theta$, a set $\mathcal{D}$ of held-out songs with timed lyrics, and a number of batches $B$, each a few windows drawn at random.
*Output:* the mean paired difference $\bar{\Delta}$ and its standard error $s$, both in nats per code, and whether the difference is clearly positive.

1 **for** $b \in [0, B)$ {
2  draw a batch of windows from the songs in $\mathcal{D}$
3  $\mathcal{L}_b^{\text{right}}$ = loss of $\theta$ with each window's own lyrics
4  $\mathcal{L}_b^{\text{none}}$ = loss of $\theta$ with empty lyrics
5  $\mathcal{L}_b^{\text{wrong}}$ = loss of $\theta$ with lines from another song
6 }
7 $\Delta_b = \mathcal{L}_b^{\text{wrong}} - \mathcal{L}_b^{\text{right}}$ for each $b$
8 $\bar{\Delta} = \frac{1}{B} \sum_b \Delta_b$
9 $s = \sqrt{\text{Var}(\Delta)/B}$
10 **return** $\bar{\Delta}$, $s$, and whether $\bar{\Delta} > 2s$

$\mathcal{L}_b$ denotes the loss of batch $b$, $\text{Var}(\Delta)$ is the sample variance of the $B$ differences, and the standard error $s$ estimates how much $\bar{\Delta}$ would vary if the test were repeated on other batches. Requiring $\bar{\Delta} > 2s$ corresponds roughly to 95 % confidence that the difference is positive. Lines 3–5 score the same windows, so the differences in line 7 compare like with like, and most of the variation between windows cancels. A model that ignores the words gives $\bar{\Delta} \approx 0$. A model that reads them should predict the audio better with the right words than with the wrong ones, giving $\bar{\Delta} > 0$, measured in nats per code. The wrong lines in line 5 always begin at a sung line rather than at silence, and comparing lines 4 and 5 helps distinguish a reaction to the presence of text from a reaction to its content.

## 7 Generation

Generation repeats a single step: the model predicts a distribution for each codebook at the next position, one code is drawn from each, and the drawn codes are appended to the sequence.

### 7.1 The key/value cache

Position $p$ attends to the keys and values of positions $0, \ldots, p$. These do not change once they have been computed, because no position attends forward, so a cache can keep them [22], and each new position then costs one pass through the blocks instead of a pass over the whole sequence. In the procedure below, $j$ indexes the blocks, $q$, $k$ and $v$ are the query, key and value vectors of position $p$, and $\pi$ is the position map of Section 6.

**DECODE-STEP**$(z, p, \mathit{cache})$

*Input:* the symbols $z$ to place at position $p$ (a byte, or the $C$ codes of a step), the position index $p$, and the cache: for every block, the key and value vectors of positions $0, \ldots, p-1$.
*Output:* for each codebook, $K$ scores for the code at position $p+1$; the cache is extended by position $p$.

1 $h = \text{embed}(z) + \text{position}(\pi(p))$
2 **for** each block $j$ {
3 $q, k, v =$ query, key and value projections of $\text{Norm}(h)$
4 $\mathit{cache}_j[p] = (k, v)$
5 $h = h + \text{Attend}(q,\ \mathit{cache}_j[0\,..\,p])$
6 $h = h + \text{FeedForward}(\text{Norm}(h))$
7 }
8 **return** the output-head scores of $\text{Norm}(h)$

**Theorem 7.1.** *Feeding a sequence position by position through* ***DECODE-STEP*** *yields, at every position, the same scores as a single forward pass over the whole sequence.*

*Proof.* We argue by induction over the blocks. At block 0, the input at position $p$ depends only on the token and the position of $p$. Suppose that the inputs of block $j$ agree at all positions up to $p$. Then the keys and values computed in line 3 agree, and when position $p$ is processed the cache holds exactly those of positions $0, \ldots, p$. In the full pass the mask gives zero weight to later positions, so attention at $p$ combines the same keys and values with the same weights, and the feed-forward network acts on each position separately. The outputs of block $j$, which are the inputs of block $j+1$, therefore agree up to position $p$. □

A test checks the theorem numerically, to within $10^{-4}$ at every position, for models with and without lyrics; the second case also checks the position map $\pi$. Because the cache has a fixed capacity, a step allocates no memory.

## 7.2 Sampling and guidance

For each codebook we keep the $k = 250$ most probable of the $K = 1\,024$ codes and draw among them in proportion to their probabilities [9]; excluding the unlikely codes avoids rare mistakes that the model could not recover from. Because the text was removed from 10 % of the training examples, the same model can predict both with the text, giving scores $z^{\text{text}}$, and without it, giving $z^{\text{none}}$; here $z$ denotes a vector of $K$ scores, which a softmax turns into probabilities. Classifier-free guidance [12] amplifies their difference:

$$z = z^{\text{none}} + s\,(z^{\text{text}} - z^{\text{none}}),$$

where $s$ is the guidance scale, a plain number ($s = 1$ uses the text as trained, and larger values follow it more strictly); this $s$ is unrelated to the standard error of Section 6.3. For instance, if two codes score $(2, 1)$ with the text and $(1, 1)$ without it, the first has probability 0.73; with $s = 3$ the scores become $(4, 1)$, and its probability rises to 0.95. We use $s = 3$, which doubles the computation per step. Lyrics longer than $L$ bytes are cut at line breaks into windows that are presented in turn, each over an equal share of the output, while the most recent audio is kept as context.

# 8 GPU kernels

Matrix products account for almost all of the arithmetic: the spectrogram, the nearest-prototype search, the projections of the Transformer and both products inside attention. Our tutorial [20] derives the shared-memory tiled product in detail; here we describe one further step, register blocking, and the way we made one gradient repeatable.

## 8.1 Register blocking

A GPU runs many *threads* at once, grouped into *blocks* that cooperate. Each thread has up to 255 fast *registers*, the threads of a block share a small on-chip *shared memory*, and all threads can reach the large but slow *global memory*. In the tiled kernel for $C = AB$, a block of $16 \times 16$ threads copies square pieces (*tiles*) of $A$ and $B$ into shared memory, which reduces traffic to global memory sixteenfold [20, 13]. Every multiply-add still reads two values from shared memory, however, and shared memory is slower than a thread's registers. Register blocking [29] lets each thread compute an $r \times r$ group of outputs, so that every value it reads is used $r$ times (Figure 6).

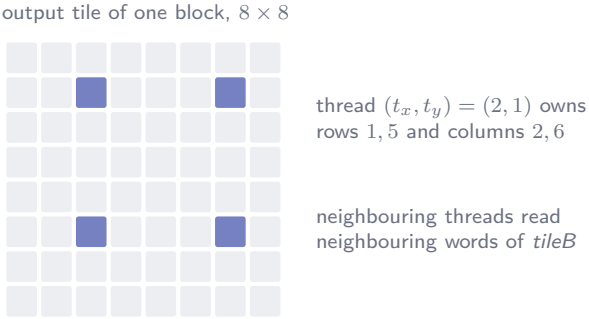



Figure 6: Register blocking with $4 \times 4$ threads and $r = 2$. Each thread computes $r^2$ outputs spread over the tile, in rows $t_y + 4i$ and columns $t_x + 4j$. Our kernel uses $16 \times 16$ threads, $r = 4$ and a $64 \times 64$ tile.

**BLOCKED-THREAD**$(t_x, t_y)$

*Input:* the column $t_x$ and row $t_y$ of a thread within a block of $16 \times 16$ threads, both in $[0, 16)$; the block computes a $64 \times 64$ tile of $C = AB$, and each thread computes $r \times r = 4 \times 4$ of its entries.
*Output:* the 16 entries of the tile owned by the thread, written to $C$.

```
1   sum[i][j] = 0 for i, j ∈ [0, r)
2   for each phase t {
3     copy a 64 × 16 tile of A into tileA
4     copy a 16 × 64 tile of B into tileB
5     BARRIER()
6     for u ∈ [0, 16) {
7       α_i = tileA[t_y + 16i][u] for i ∈ [0, r)
8       β_j = tileB[u][t_x + 16j] for j ∈ [0, r)
9       sum[i][j] = sum[i][j] + α_i β_j for i, j ∈ [0, r)
10    }
11    BARRIER()
12  }
13  store sum[i][j] at row t_y + 16i, column t_x + 16j of the tile
```

**BARRIER**() waits until every thread of the block has reached it. A *phase* handles 16 consecutive values of the inner dimension, so the loop of line 2 runs once for every 16 columns of $A$. *tileA* and *tileB* live in shared memory, while *sum*, $\alpha$ and $\beta$ live in the thread's registers; in this subsection $\alpha$ and $\beta$ are these register copies, unrelated to the momentum and the Kaiser parameter above. Lines 7 and 8 read $2r = 8$ values from shared memory, and line 9 performs $r^2 = 16$ multiply-adds, so each multiply-add costs $2/r = 0.5$ reads instead of 2. Global traffic falls as well: each phase copies $2 \cdot 64 \cdot 16$ values for $64 \cdot 64 \cdot 16$ multiply-adds, so the product of an $M \times K$ and a $K \times N$ matrix (here $M$, $N$ and $K$ are matrix dimensions) fetches

$$\frac{M}{64} \cdot \frac{N}{64} \cdot \frac{K}{16} \cdot 2 \cdot 64 \cdot 16 = \frac{2MNK}{64}$$

values of 4 bytes each, a quarter of the $2MNK/16$ needed with $16 \times 16$ tiles. The two barriers play the same roles as in the tiled kernel, whose correctness is proved in [20]. On the Orin NX the blocked kernel is 1.8 to 2.3 times faster than the tiled one (Table 7).

## 8.2 A repeatable embedding gradient

The embedding table has one row of $d$ numbers per vocabulary symbol, and its gradient adds the gradient vector $g_p$ (also $d$ numbers) of each position $p$ into the row of the token $z_p$ that the position holds. Many positions share a token. With atomic additions, threads would add into a shared row in an order that varies from run to run, and because floating-point addition is not associative, the result would vary as well. We fix the order with a counting sort.

**EMBEDDING-GRADIENT**$(z, g, V)$

*Input:* for every position $p$, its token $z_p \in [0, V)$ and its gradient vector $g_p$ of $d$ numbers; and the number of rows $V$ of the embedding table (the vocabulary size).
*Output:* the gradient $G$ of the embedding table, a $V \times d$ matrix with one row per token; $c \in [0, d)$ indexes its columns.

```
count[v] = |{p : z_p = v}| for v ∈ [0, V)  // on the CPU
start[v] = Σ_{u<v} count[u]
list = positions sorted by token, then by position
for each token v and column c, in parallel {  // on the GPU
    G[v][c] = Σ g_p[c] over p in list[start[v] .. start[v + 1]), in order
}
```

For the tokens $z = (2, 0, 2, 1)$, the counts are $(1, 1, 2)$, the starts are $(0, 1, 2, 4)$ and the list is $(1, 3, 0, 2)$, so token 2 receives $g_0 + g_2$ in that order. Each entry is summed by one thread in a fixed order, so the result is the same on every run, and a whole training run can be repeated bit for bit on the same device.

# 9 Experiments

The hardware is described in Table 6 in the appendix. The main corpus is the Song Describer Dataset [18], 706 two-minute Creative Commons excerpts with 1 106 captions written by listeners. Songs are divided between training and evaluation by artist, so that no voice appears on both sides, which leaves 667 songs for training and 39 for evaluation. We report spectral errors as the mean absolute difference between two log-mel spectrograms, averaged over frames and bands, in *log units per band*. Because the spectrogram holds natural logarithms of energy, an error of one log unit means that a band's energy is off by a factor of $e \approx 2.7$, or $10 \log_{10} e \approx 4.3$ dB.

**Correctness and speed.** All 38 tests pass on the Orin NX, and the kernel tests also pass on the Thor. They compare every kernel with a CPU implementation at deliberately awkward sizes, check every gradient numerically, verify the cache theorem, and confirm that a small network can memorise a fixed batch. On the Orin NX, register blocking makes a training step of the first model 1.8 times faster than tiling and 9.3 times faster than the direct kernel (Table 8).

**First training runs.** Table 2 summarises the runs with the first tokenizer. A training *step* here is one update of the parameters, computed from a batch of 8 sequences; the held-out loss is measured on songs by artists that the model never heard during training. A run of 1 500 steps reached a held-out loss of 4.93, and a run of 15 000 steps on the Thor reached 4.02. When we restricted the data to the 281 songs whose captions mention a voice, a model fine-tuned from the longer run reached its best held-out loss of 4.23 after only 250 steps, after which its held-out loss rose while its training loss continued to fall; a model trained from scratch on the same songs reached 4.72. The vocal subset appears to be too small for long training.

## 9.1 What limits the sound

The generated voices sounded pitched but blurred. To find out why, without involving the model, we rendered 12-second excerpts of four held-out vocal songs through each stage of the pipeline and compared the results with their exact spectrograms (Figure 7 and Table 3).

Table 2: Best held-out loss with the first tokenizer, in nats per code (uniform guess: 6.93).

| run | data | steps | loss (nats per code) |
|---|---|---|---|
| first run, Orin NX | all songs | 1 500 | 4.93 |
| longer run, Thor | all songs | 15 000 | 4.02 |
| fine-tuned, Thor | vocal songs | 250 | 4.23 |
| from scratch, Orin NX | vocal songs | 5 500 | 4.72 |

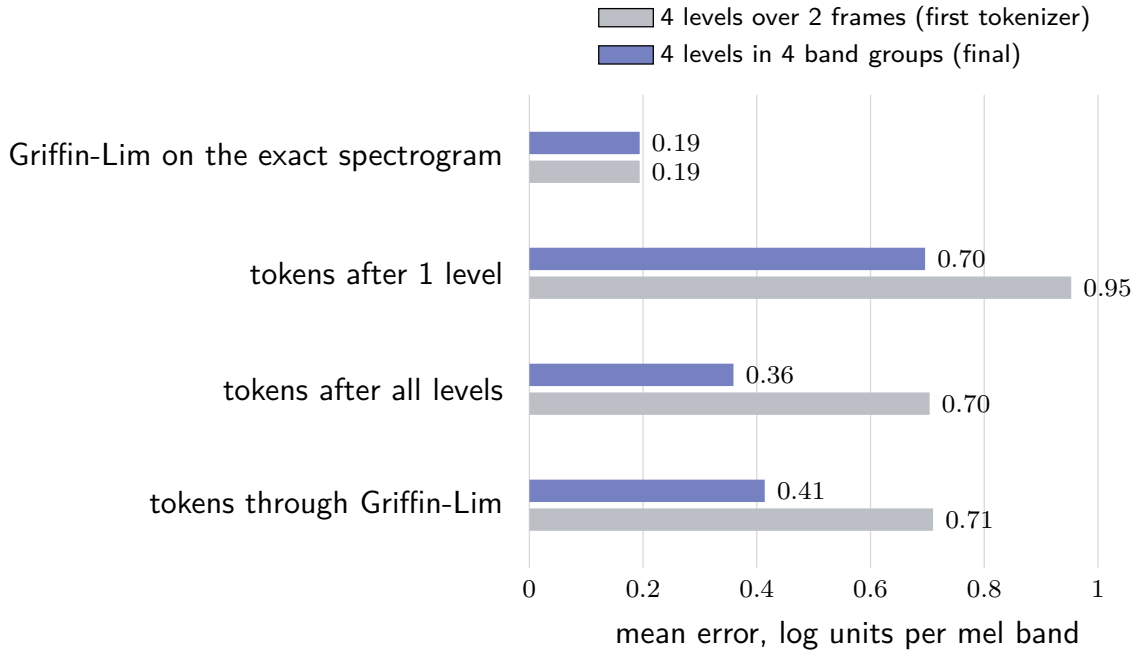



Figure 7: Mean spectral error of each stage on four held-out vocal excerpts, for the first tokenizer (grey) and the final one (blue). The bottom pair is the best that any model trained on these tokens could achieve.

Table 3: Spectral error of each stage on four held-out vocal excerpts, in log units per mel band, for the first tokenizer and the final one.

| | | first tokenizer | | final tokenizer | |
|---|---|---|---|---|---|
| excerpt | vocoder | tokens | end to end | tokens | end to end |
| guitar, brass, female voice | 0.230 | 0.807 | 0.821 | 0.425 | 0.480 |
| folk metal, male voice | 0.152 | 0.707 | 0.703 | 0.353 | 0.378 |
| electronic, vocal sample | 0.170 | 0.569 | 0.554 | 0.268 | 0.338 |
| bass, piano, wordless voice | 0.222 | 0.732 | 0.763 | 0.390 | 0.460 |
| mean | 0.194 | 0.704 | 0.710 | 0.359 | 0.414 |

With the first tokenizer, the tokens alone caused 3.6 times the error of the vocoder, and the end-to-end error of a perfect model, 0.71, was almost entirely due to the tokenizer. An error of about 0.7 log units, roughly 3 dB in every band, is enough to blur the harmonic and formant detail on which an intelligible voice depends. This measure does not capture the phase artefacts of Griffin-Lim, which remain audible, so the tokenizer is the main spectral limitation rather than the only audible one.

To examine the model on its own, we trained it on a single 12-second excerpt of 562 steps. Its training loss fell to 0.054 after 100 steps and to 0.002 after 200. We then generated 12 seconds from the excerpt's caption and compared the generated codes with the excerpt's own (Table 4). Greedy decoding reproduced all 562 steps, and top-$k$ sampling departed from the excerpt only at isolated codes, without drifting away.

Table 4: Generated codes compared with a memorised excerpt.

| decoding | agreement per codebook (%) | identical prefix (steps) |
|---|---|---|
| greedy | 100.0, 100.0, 100.0, 100.0 | 562 of 562 |
| top-$k$, seed 1 | 99.6, 100.0, 100.0, 99.6 | 26 |
| top-$k$, seed 2 | 100.0, 99.8, 99.3, 99.8 | 26 |

The model can reproduce audio exactly, and when it does, the result sounds like the tokenizer's reconstruction. These two observations led us to improve the tokens before training larger models.

### 9.2 A finer tokenizer

Guided by this analysis, we compared tokenizers that differ in the number of frames per step, the number of band groups and the number of levels (Table 10). Quantising one frame per step in 4 band groups with 4 levels each lowered the mean error on the held-out songs from 0.776 to 0.401. A control without band groups suggests that most of this gain comes from the larger number of codes per step, and a smaller part from the grouping. On the four vocal excerpts of Figure 7, the error of the tokens fell from 0.70 to 0.36, and the end-to-end error from 0.71 to 0.41. The price is a vocabulary of 16 codebooks instead of 4.

### 9.3 A larger model on the finer tokens

With the finer tokens we began training a larger model on the Thor: 8 blocks of width 512, 42.9 million parameters, with the lyrics input enabled from the start. It is trained on all 742 training songs of the combined corpora, with the lyrics corpus weighted 4-fold and two takes recorded by the first author weighted 10-fold, in batches of 8 sequences of 1 408 positions. A step takes 2.0 seconds and about 11 GB of the Thor's 128 GB of memory. Training is still in progress; after 8 000 of 20 000 steps the held-out loss is 5.11 nats per code, averaged over its 16 codebooks, and still falling. Because the vocabulary differs, this figure is not comparable with those of Table 2.

### 9.4 Lyrics

We extended both vocal models of Table 2 with a lyrics field and trained them further on vocal songs, JamendoLyrics, and songs whose captions mention rock or a related style. Table 5 gives the result of **Lyrics-Test** for a model trained on the Thor with the lyrics corpus weighted 8-fold, over 48 batches of 4 windows.

Table 5: Loss with the right lyrics, no lyrics and the wrong lyrics, and the paired difference $\bar{\Delta} \pm s$.

| songs | right | none | wrong | $\bar{\Delta} \pm s$ |
|---|---|---|---|---|
| | | (nats per code) | | |
| held out (6) | 4.3345 | 4.3409 | 4.3345 | $+0.0000 \pm 0.0004$ |
| training (73) | 4.0337 | 4.0395 | 4.0378 | $+0.0042 \pm 0.0006$ |

On the held-out songs, the right and the wrong lyrics give the same loss, so this model does not yet use the words. On the training songs the difference is statistically clear but small, which we take to indicate memorisation of those songs rather than a learned relation between words and sound. We hope that the paired test will be useful as a clear criterion for future attempts.

## 10 Limitations

The tokenizer, although improved, still limits the fidelity of the output, and Griffin-Lim adds phase artefacts that a learned vocoder would remove. The data contain comparatively little singing (298 of the 706 songs mention a voice) and very little of some styles (two songs are described as grunge). Our models are much smaller than published systems [3], and we have not yet carried out a formal listening test, so we make no claims about perceived quality. On the implementation side, all arithmetic uses 32-bit floating point without tensor cores, and attention stores its full matrix of scores [5].

## 11 Conclusion

We have described StaleyNET, a complete text-to-music system written in Rust and trained from scratch on embedded GPUs. Building every stage ourselves allowed us to measure each one separately, and these measurements pointed to the tokenizer, rather than the language model, as the main limitation on sound quality. A revised tokenizer that quantises groups of frequency bands roughly halved the reconstruction error, and a larger model is now being trained on it. We hope that the worked examples, the stage-by-stage analysis and the paired lyrics test will be of use to others who build similar systems on modest hardware.

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# A Hardware and software

Both devices were shared with other work during our experiments. In particular, language-model services occupied most of the Thor's memory until the larger training run began, at which point they were stopped.

Table 6: The two devices. Core counts and memory bandwidth are the manufacturer's specifications; memory capacity, compute capability and software versions were read from the devices.

| | Jetson Orin NX 16 GB | Jetson AGX Thor |
|---|---|---|
| GPU | NVIDIA Ampere, 1 024 CUDA cores, 32 tensor cores | NVIDIA Blackwell, 2 560 CUDA cores, 96 tensor cores |
| compute capability | 8.7 | 11.0 |
| CPU | 8-core Arm Cortex-A78AE | 14-core Arm Neoverse-V3AE |
| memory | 16 GB LPDDR5, shared by CPU and GPU | 128 GB LPDDR5X, shared by CPU and GPU |
| memory bandwidth | 102 GB/s | 273 GB/s |
| system software | L4T R36.4.3, CUDA 12.6 | L4T R39.2.1, CUDA 13.2 |
| Rust toolchain | nightly-2026-04-03 | nightly-2026-10-11 |
| cuda-oxide | revision b7fe543 | revision e027ac3 |

# B Benchmarks

All figures in this appendix were measured on the Orin NX with `staleynet bench`. Each timing is the mean of five runs after one warm-up run, and each table entry is the median of three such measurements, because the clock frequency of the device varies noticeably from run to run.

Table 7: Matrix products at the sizes used by the first model, in GFLOP/s.

| product ($M \times N \times K$) | direct | tiled | register-blocked | blocked / tiled |
|---|---|---|---|---|
| attention projections ($2048 \times 768 \times 256$) | 46.8 | 258.6 | 579.0 | 2.2 |
| feed-forward ($2048 \times 1024 \times 256$) | 55.3 | 255.5 | 450.6 | 1.8 |
| output heads ($2048 \times 4096 \times 256$) | 31.2 | 222.1 | 458.5 | 2.1 |
| spectrogram ($4096 \times 513 \times 1024$) | 47.7 | 236.1 | 541.7 | 2.3 |

Table 8: One training step of the first model on $4 \times 512$ positions, forward and backward.

| matrix-product kernel | time per step (ms) |
|---|---|
| direct | 2 930 |
| tiled | 576 |
| register-blocked | 315 |

On the Thor, measured during training runs of identical shape, register blocking reduced a training step of the first model on $4 \times 768$ positions from 238 ms to 147 ms, and a step of the larger model on $8 \times 1\,408$ positions takes about 2.0 seconds.

Table 9: Other measurements on the Orin NX.

| measurement | result |
|---|---|
| log-mel spectrogram of 60 s, GPU | 47 ms |
| log-mel spectrogram of 60 s, CPU (rustfft, 8 threads) | 73 ms |
| decoding step, key/value cache | 6.2 ms |
| decoding step, recomputing the context | 17.1 ms |

# C Tokenizer study

Table 10 lists the mean reconstruction error over all 39 held-out songs for the tokenizers we compared, each trained on the same training songs with 1 024 prototypes per codebook. The columns give the number of frames per step, the number of band groups, the number of residual levels in each group, and the resulting number of codes per step.

Table 10: Held-out reconstruction error of different tokenizers, in log units per mel band (one log unit is a factor of $e$ in energy, about 4.3 dB). A frame covers 10.7 ms of audio; every codebook has 1 024 prototypes.

| frames | groups | levels | codes | error (log units per band) |
|---|---|---|---|---|
| 2 | 1 | 4 | 4 | 0.776 |
| 2 | 1 | 8 | 8 | 0.643 |
| 1 | 1 | 4 | 4 | 0.745 |
| 1 | 1 | 8 | 8 | 0.588 |
| 1 | 4 | 2 | 8 | 0.597 |
| 1 | 8 | 1 | 8 | 0.612 |
| 1 | 1 | 16 | 16 | 0.427 |
| 1 | 8 | 2 | 16 | 0.403 |
| 1 | 4 | 4 | 16 | 0.401 |

At 8 codes per step, dividing the bands into groups did not help. With 16 codes per step it helped a little: 16 levels without groups reached 0.427, and 4 groups of 4 levels reached 0.401. Most of the improvement over the first tokenizer therefore comes from describing each step with more codes, and a smaller part, about 6 %, from quantising the band groups separately.